



Factoring Polynomials: GCF, Grouping, and Difference of Two Squares

Intermediate Algebra Worksheet · Grade 9-12 · numberbender.com

Name: _____

Date: _____

Score: / 10

Learning Objectives

- Find the greatest common factor (GCF) of a polynomial
- Factor polynomials by grouping
- Recognize and factor the difference of two squares
- Apply factoring techniques to simplify polynomial expressions

Factor each polynomial completely using the appropriate technique (GCF, grouping, or difference of two squares).

1. Find the GCF of the terms.

$$12x^3, 18x^2, 24x$$

Answer: _____

2. Factor out the GCF.

$$15x^4 - 25x^3 + 10x^2$$

Answer: _____

3. Factor out the GCF.

$$8a^3b^2 - 12a^2b^3 + 20ab^2$$

Answer: _____

4. Factor by grouping.

$$x^3 + 2x^2 + 3x + 6$$

Answer: _____

5. Factor by grouping.

$$6x^2 + 9x + 4x + 6$$

Answer: _____

6. Factor by grouping.

$$2x^3 - 6x^2 + 5x - 15$$

Answer: _____

7. Factor the difference of two squares.

$$x^2 - 49$$

Answer: _____



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Factoring Polynomials GCF Grouping and Difference of Two Squares Work...

More free worksheets, video lessons & courses:
numberbender.com/math-worksheets

Page 1 of 4 · Video: youtu.be/54Wgb1R75vw



Watch the lesson



More free sheets

8. Factor the difference of two squares.

$$9y^2 - 25$$

Answer: _____

9. Factor completely (use GCF first, then difference of squares).

$$3x^3 - 27x$$

Answer: _____

10. Factor completely.

$$16x^4 - 81$$

Answer: _____





Factoring Polynomials: GCF, Grouping, and Difference of Two Squares

ANSWER KEY & SOLUTIONS

Intermediate Algebra Worksheet · Grade 9-12 · numberbender.com

This worksheet covers all topics from the video: warm-up review, GCF factoring, factoring by grouping, and difference of two squares. Problems progress from basic GCF identification through multi-step grouping and special pattern recognition.

Solutions

1. Find the GCF of the terms.

$$12x^3, 18x^2, 24x$$

- The greatest common numerical factor of 12, 18, and 24 is 6.
- The smallest power of x common to all terms is x to the first power.
- Multiply these together to get the GCF of $6x$.

Answer: $6x$

2. Factor out the GCF.

$$15x^4 - 25x^3 + 10x^2$$

- Identify the GCF of the coefficients 15, 25, and 10, which is 5.
- The smallest power of x in all terms is x squared.
- Factor out $5x$ squared from each term to get $5x$ squared times the quantity $3x$ squared minus $5x$ plus 2.

Answer: $5x^2(3x^2 - 5x + 2)$

3. Factor out the GCF.

$$8a^3b^2 - 12a^2b^3 + 20ab^2$$

- The GCF of 8, 12, and 20 is 4.
- The smallest power of a is a to the first, and the smallest power of b is b squared.
- Factor out $4ab$ squared from each term to get $4ab$ squared times the quantity $2a$ squared minus $3ab$ plus 5.

Answer: $4ab^2(2a^2 - 3ab + 5)$

4. Factor by grouping.

$$x^3 + 2x^2 + 3x + 6$$

- Group the first two terms and the last two terms.
- Factor x squared from the first group and 3 from the second group to get x squared times the quantity x plus 2 plus 3 times the quantity x plus 2.
- The common binomial factor is x plus 2, so the result is the quantity x plus 2 times the quantity x squared plus 3.

Answer: $(x + 2)(x^2 + 3)$

5. Factor by grouping.

$$6x^2 + 9x + 4x + 6$$

- Group the first two terms and the last two terms.
- Factor $3x$ from the first group and 2 from the second group to get $3x$ times the quantity $2x$ plus 3 plus 2 times the quantity $2x$ plus 3.
- Factor out the common binomial $2x$ plus 3 to get the quantity $3x$ plus 2 times the quantity $2x$ plus 3.

Answer: $(3x + 2)(2x + 3)$



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Factoring Polynomials GCF Grouping and Difference of Two Squares Work...

More free worksheets, video lessons & courses:

numberbender.com/math-worksheets

Page 3 of 4 · Video: youtu.be/54Wgb1R75vw



Watch the lesson



More free sheets

6. Factor by grouping.

$$2x^3 - 6x^2 + 5x - 15$$

→ Group the first two terms and the last two terms.

→ Factor $2x$ squared from the first group and 5 from the second group to get $2x$ squared times the quantity x minus 3 plus 5 times the quantity x minus 3.

→ The common binomial factor is x minus 3, giving the quantity x minus 3 times the quantity $2x$ squared plus 5.

Answer: $(x - 3)(2x^2 + 5)$

7. Factor the difference of two squares.

$$x^2 - 49$$

→ Recognize that x squared and 49 are both perfect squares.

→ The square root of x squared is x and the square root of 49 is 7.

→ Apply the difference of squares pattern to write the factored form as the quantity x plus 7 times the quantity x minus 7.

Answer: $(x + 7)(x - 7)$

8. Factor the difference of two squares.

$$9y^2 - 25$$

→ Identify $9y$ squared as the square of $3y$ and 25 as the square of 5.

→ Apply the difference of squares formula a squared minus b squared equals the quantity a plus b times the quantity a minus b .

→ Write the factored form as the quantity $3y$ plus 5 times the quantity $3y$ minus 5.

Answer: $(3y + 5)(3y - 5)$

9. Factor completely (use GCF first, then difference of squares).

$$3x^3 - 27x$$

→ Factor out the GCF of $3x$ from both terms to get $3x$ times the quantity x squared minus 9.

→ Recognize x squared minus 9 as a difference of two squares.

→ Factor it as the quantity x plus 3 times the quantity x minus 3, giving $3x$ times the quantity x plus 3 times the quantity x minus 3.

Answer: $3x(x + 3)(x - 3)$

10. Factor completely.

$$16x^4 - 81$$

→ Recognize $16x$ to the fourth and 81 as perfect squares, so apply the difference of squares to get the quantity $4x$ squared plus 9 times the quantity $4x$ squared minus 9.

→ The factor $4x$ squared minus 9 is itself a difference of squares.

→ Factor it as the quantity $2x$ plus 3 times the quantity $2x$ minus 3 to get the final answer of the quantity $4x$ squared plus 9 times the quantity $2x$ plus 3 times the quantity $2x$ minus 3.

Answer: $(4x^2 + 9)(2x + 3)(2x - 3)$

